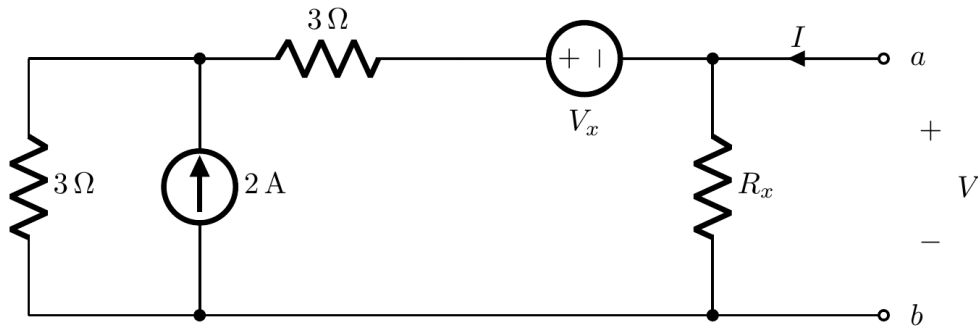


- (a) Draw the equivalent circuit using the Thevenin theorem. [3.5 marks]
 (b) Find the Norton equivalent current. [1.5 marks]

Question 3.

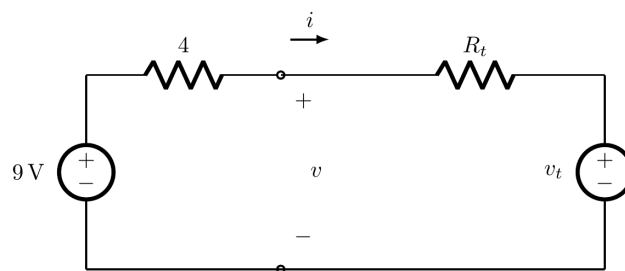
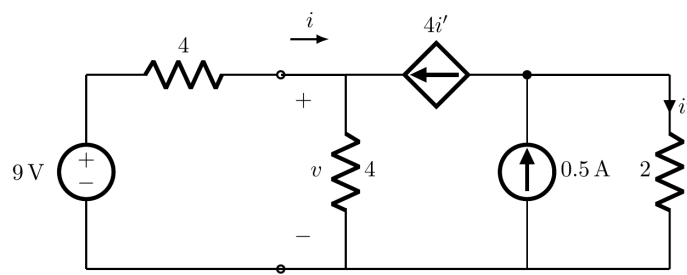
In the circuit below, all resistors are practical and cannot have negative values. When a voltage of $V=2\text{ V}$ is applied across terminals a and b , the circuit draws a current of $I=3\text{ A}$. However, if -6 V is applied, instead of drawing, the circuit supplies 1 A to terminal a .



- (a) Determine the unknown resistance R_x in the circuit. [4 marks]
 (b) Find an equivalent representation of the circuit having the minimum number of elements, and determine V_x . [6 marks]

Question 4.

Determine the values of R_t and v_t , if the following two circuits are equivalent to each other.



- (i) Determine the values of $R_{\square}$ and $v_{\square}$. **[2.5 marks]**
- (ii) Plot the IV relation of the blue part and find the resistance of the dependent source. **[2.5 marks]**